JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2025 2nd April Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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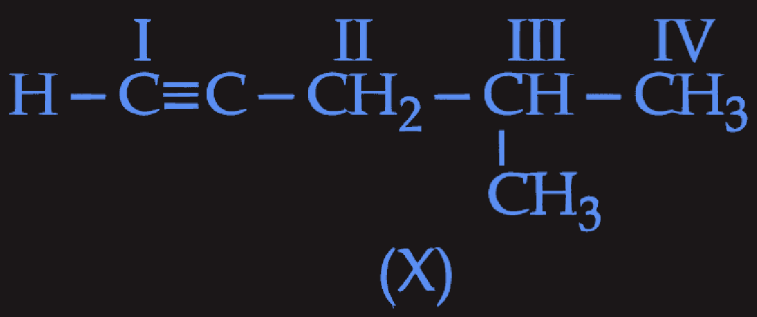
QR codes open the DekhoExams website.

Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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Consider the following compound (X)



The most stable and least stable carbon radicals, respectively, produced by homolytic cleavage of corresponding $C - H$ bond are :

(A) III, II (B) II, IV (C) I, IV (D) II, I

Q2. MCQ (Single Correct Answer)

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$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Consider the above reaction, what mass of $CaCl_2$ will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of $CaCO_3$?

(Given : Molar mass of Ca, C, O, H and Cl are $40, 12, 16, 1$ and $35.5\ g\ mol^{-1}$, respectively)

(A) 3.908 g (B) 2.636 g (C) 10.545 g (D) 5.272 g

Q3. MCQ (Single Correct Answer)

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If equal volumes of AB_2 and XY (both are salts) aqueous solutions are mixed, which of the following combination will give a precipitate of AY_2 at 300 K ? (Given K_{sp} (at 300 K) for $AY_2 = 5.2 \times 10^{-7}$)

(A) $2.0 \times 10^{-4} MAB_2, 0.8 \times 10^{-3} MXY$

(B) $2.0 \times 10^{-2} MAB_2, 2.0 \times 10^{-2} MXY$

(C) $1.5 \times 10^{-4} MAB_2, 1.5 \times 10^{-3} MXY$

(D) $3.6 \times 10^{-3} MAB_2, 5.0 \times 10^{-4} MXY$

Q4. MCQ (Single Correct Answer)

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Given below are two statements :

Statement (I) : The metallic radius of Al is less than that of Ga .

Statement (II) : The ionic radius of Al^{3+} is less than that of Ga^{3+}.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are correct

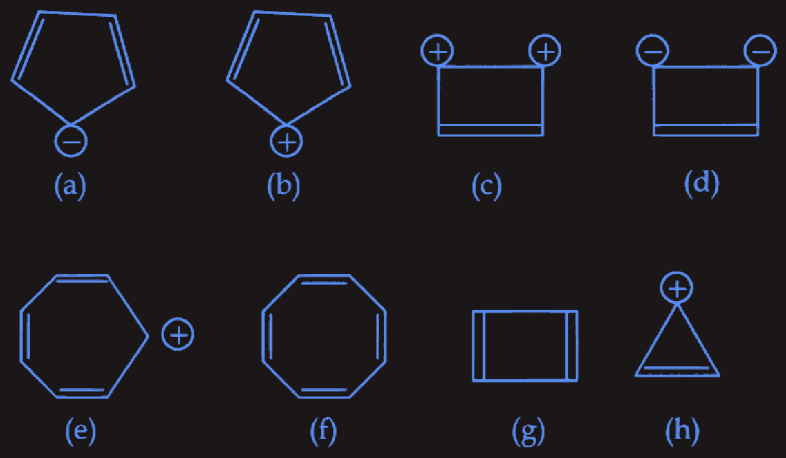
(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are incorrect

Q5. MCQ (Single Correct Answer)

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Designate whether each of the following compounds is aromatic or not aromatic.



(A) a, c, d, e, h aromatic and b, f, g not aromatic

(B) b, e, f, g aromatic and a, c, d, h not aromatic

(C) a, b, c, d aromatic and e, f, g, h not aromatic

(D) e, g aromatic and a, b, c, d, f, h not aromatic

Q6. MCQ (Single Correct Answer)

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Choose the correct tests with respective observations.

(A) $CuSO_4$ (acidified with acetic acid) $+K_4[Fe(CN)_6] \rightarrow$ Chocolate brown precipitate.

(B) $FeCl_3 + K_4[Fe(CN)_6] \rightarrow$ Prussian blue precipitate.

(C) $ZnCl_2 + K_4[Fe(CN)_6]$, neutralised with $NH_4OH \rightarrow$ White or bluish white precipitate.

(D) $MgCl_2 + K_4[Fe(CN)_6] \rightarrow$ Blue precipitate.

(E) $BaCl_2 + K_4[Fe(CN)_6]$, neutralised with $NaOH \rightarrow$ White precipitate.

Choose the correct answer from the options given below :

(A) A, D and E only (B) A, B and C only (C) B, D and E only (D) C, D and E only

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Q7. MCQ (Single Correct Answer)

A solution is made by mixing one mole of volatile liquid A with 3 moles of volatile liquid B. The vapour pressure of pure A is 200 mm Hg and that of the solution is 500 mm Hg . The vapour pressure of pure B and the least volatile component of the solution, respectively, are:

(A) $1400\ \mathrm{mm\ Hg}, \mathrm{A}$

(B) $1400\ \mathrm{mm\ Hg}, B$

(C) $600\ \mathrm{mm\ Hg}, \mathrm{A}$

(D) $600\ \mathrm{mm\ Hg}, \mathrm{B}$

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Q8. MCQ (Single Correct Answer)

A molecule with the formula $\mathrm{AX_4Y}$ has all its elements from p-block. Element A is rarest, monoatomic, non-radioactive from its group and has the lowest ionization enthalpy value among A, X and Y . Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is:

(A) Pentagonal planar

(B) Square pyramidal

(C) Trigonal bipyramidal

(D) Octahedral

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Q9. MCQ (Single Correct Answer)

Given below are two statements :

Statement (I) : In octahedral complexes, when $\Delta_0 < \mathrm{P}$ high spin complexes are formed. When $\Delta_0 > P$ low spin complexes are formed.

Statement (II) : In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

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Q10. MCQ (Single Correct Answer)

On complete combustion 1.0 g of an organic compound (X) gave 1.46 g of $\mathrm{CO_2}$ and 0.567 g of $\mathrm{H_2O}$. The empirical formula mass of compound (X) is _____ g. (Given molar mass in $\mathrm{gmol^{-1}C : 12, H : 1, O : 16}$)

(A) 60

(B) 45

(C) 30

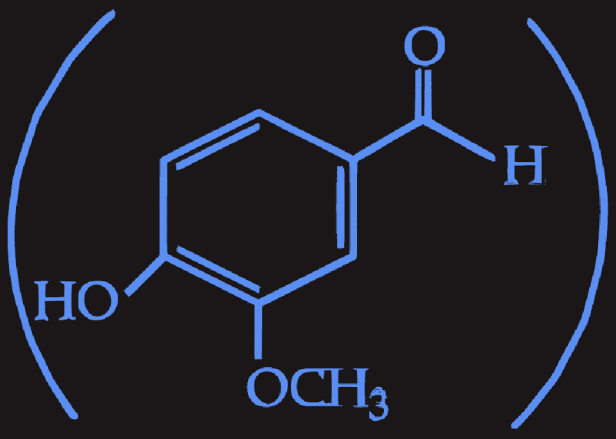
(D) 15

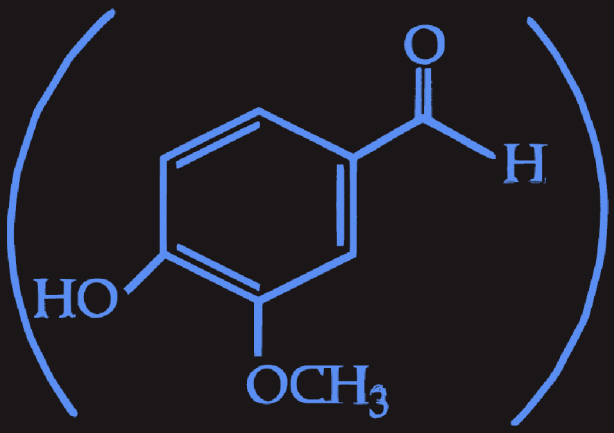
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Q11. MCQ (Single Correct Answer)

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Given below are two statements:

Statement I : Vanilin  will react with NaOH and also with Tollen's reagent.

Statement II : Vanilin  will undergo self aldol condensation very easily.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Q12. MCQ (Single Correct Answer)

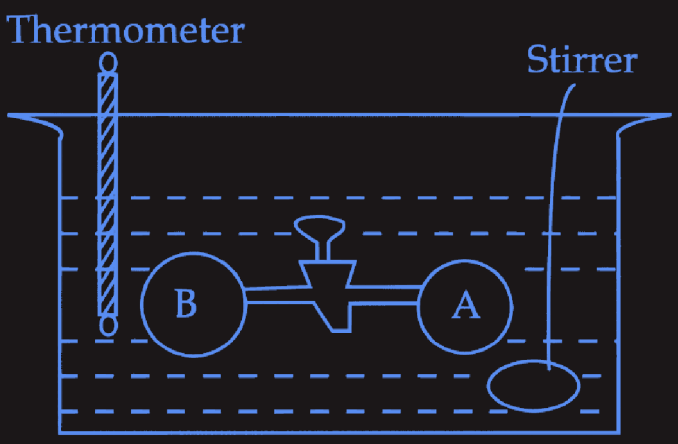
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An optically active alkyl halide $C_4H_9Br[A]$ reacts with hot KOH dissolved in ethanol and forms alkene $[B]$ as major product which reacts with bromine to give dibromide $[C]$. The compound [C] is converted into a gas [D] upon reacting with alcoholic $NaNH_2$. During hydration 18 gram of water is added to 1 mole of gas [D] on warming with mercuric sulphate and dilute acid at 333 K to form compound [E]. The IUPAC name of compound [E] is :

(A) Butan-2-ol (B) But-2-yne (C) Butan-2-one (D) Butan-1-al

Q13. MCQ (Single Correct Answer)

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Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true ?

(A) $dq \neq 0$

(B) The pressure in the vessel B before opening the stopcock is zero

(C) $dw \neq 0$

(D) $dU \neq 0$

Q14. MCQ (Single Correct Answer)

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Among $SO_2, NF_3, NH_3, XeF_2, ClF_3$ and SF_4, the hybridization of the molecule with nonzero dipole moment and highest number of lone-pairs of electrons on the central atom is

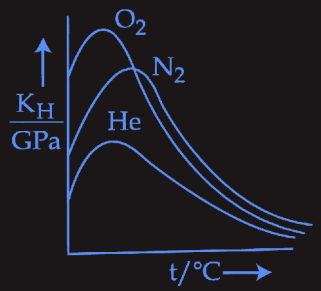
(A) sp^3 (B) dsp^2 (C) sp^3d^2 (D) sp^3d

Q15. MCQ (Single Correct Answer)

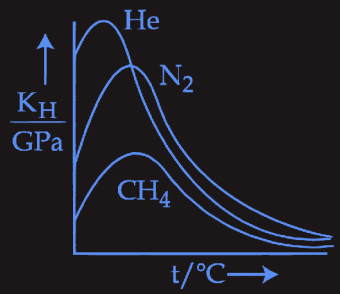
Which of the following graph correctly represents the plots of K_H at 1 bar for gases in water versus temperature?

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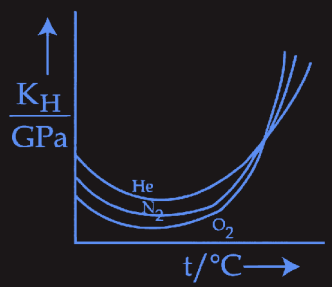
(A)



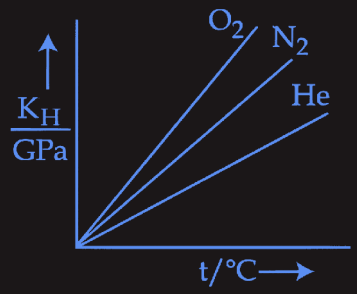
(B)



(C)



(D)



Q16. MCQ (Single Correct Answer)

According to Bohr's model of hydrogen atom, which of the following statement is incorrect?

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(A) Radius of 4th orbit is four times larger than that of 2nd orbit

(B) Radius of 8th orbit is four times larger than that of 4th orbit

(C) Radius of 6th orbit is three times larger than that of 4th orbit

(D) Radius of 3rd orbit is nine times larger than that of 1st orbit

Q17. MCQ (Single Correct Answer)

Identify the correct statement among the following :

(A) All naturally occurring amino acids except glycine contain one chiral centre.

(B) Amino acid, cysteine can easily undergo dimerisation due to the presence of free SH group.

(C) Glutamic acid is the only amino acid that contains a $-COOH$ group at the side chain.

(D) All naturally occurring amino acids are optically active.

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Q18. MCQ (Single Correct Answer)

The correct order of basic nature in aqueous solution for the bases

$NH_3, H_2N-NH_2, CH_3CH_2NH_2, (CH_3CH_2)_2NH$ and $(CH_3CH_2)_3N$ is :

(A) $NH_3 < H_2N-NH_2 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$

(B) $H_2N-NH_2 < NH_3 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$

(C) $NH_2-NH_2 < NH_3 < CH_3CH_2NH_2 < (CH_3CH_2)_3N < (CH_3CH_2)_2NH$

(D) $NH_3 < H_2N-NH_2 < CH_3CH_2NH_2 < (CH_3CH_2)_2NH < (CH_3CH_2)_3N$

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Q19. MCQ (Single Correct Answer)

The property/properties that show irregularity in first four elements of group-17 is/are

(A) Covalent radius

(B) Electron affinity

(C) Ionic radius

(D) First ionization energy

Choose the correct answer from the options given below :

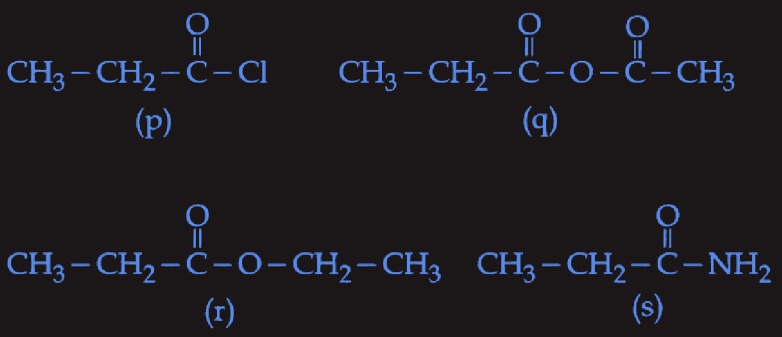
(A) A, B, C and D (B) A and C only (C) B only (D) B and D only

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Q20. MCQ (Single Correct Answer)

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Consider the following molecules:



The correct order of rate of hydrolysis is :

(A) $\mathrm{p} > \mathrm{q} > \mathrm{r} > \mathrm{s}$

(B) $r > q > p > s$

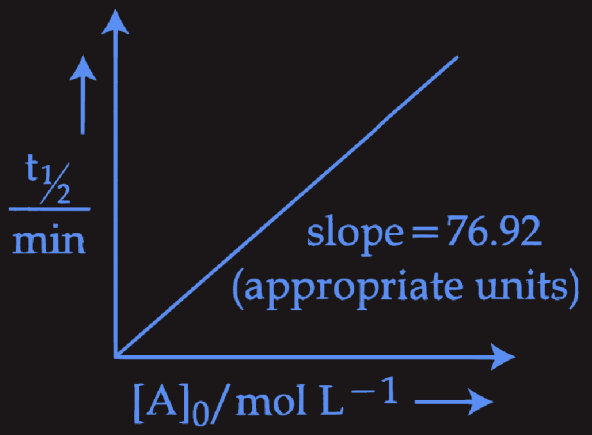
(C) $\mathrm{p} > \mathrm{r} > \mathrm{q} > \mathrm{s}$

(D) $q > p > r > s$

Q21. Numerical

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For the reaction A $\rightarrow$ products.



The concentration of A at 10 minutes is _____ $\times 10^{-3}\ \mathrm{mol\,L^{-1}}$ (nearest integer). The reaction was started with $2.5\ \mathrm{mol\,L^{-1}}$ of A .

Q22. Numerical

0.1 mol of the following given antiviral compound (P) will weigh _____ $\times 10^{-1}$ g (nearest integer).



(P)

(Given : molar mass in gmol^{-1} H : 1, C : 12, N : 14, O : 16, F : 19, I : 127)

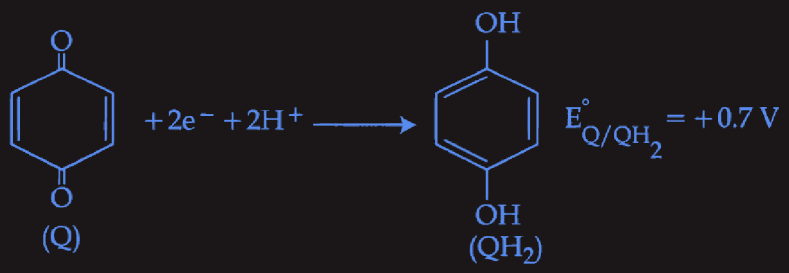
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Q23. Numerical

Consider the following electrochemical cell at standard condition.

$\mathrm{Au(s)|QH_2, Q|NH_4X(0.01M)||Ag^+(1M)\ |\ Ag(s)}\ \mathrm{E_{cell}} = +0.4\ \mathrm{V}$

The couple $\mathrm{QH_2/Q}$ represents quinhydrone electrode, the half cell reaction is given below:



$\left[\text{Given : } \mathrm{E^0_{Ag^+/Ag}} = +0.8\ \mathrm{V} \text{ and } \frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.06\ \mathrm{V}\right]$

The $\mathrm{pK_b}$ value of the ammonium halide salt $(\mathrm{NH_4X})$ used here is _____ . (nearest integer)

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Q24. Numerical

Consider the following equilibrium,

$\mathrm{CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)}$

0.1 mol of CO along with a catalyst is present in a $2\mathrm{dm}^3$ flask maintained at 500 K . Hydrogen is introduced into the flask until the pressure is 5 bar and 0.04 mol of $\mathrm{CH_3OH}$ is formed. The K_p^θ is _____ $\times 10^{-3}$ (nearest integer).

Given : $\mathrm{R} = 0.08\ \mathrm{dm}^3$ bar $\mathrm{K}^{-1}\ \mathrm{mol}^{-1}$

Assume only methanol is formed as the product and the system follows ideal gas behaviour.

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Q25. Numerical

A transition metal (M) among $\mathrm{Mn}, \mathrm{Cr}, \mathrm{Co}$ and Fe has the highest standard electrode potential $\left(\mathrm{M}^{3+} / \mathrm{M}^{2+}\right)$. It forms a metal complex of the type $\left[\mathrm{M}(\mathrm{CN})_6\right]^{4-}$. The number of electrons present in the $\mathrm{e_g}$ orbital of the complex is _____.

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let $\mathrm{A}=\begin{bmatrix}\alpha & -1 \\ 6 & \beta\end{bmatrix}, \alpha>0$, such that $\det(\mathrm{A})=0$ and $\alpha+\beta=1$. If I denotes 2×2 identity matrix, then the matrix $(I+A)^8$ is :

(A) $\begin{bmatrix}257 & -64 \\ 514 & -127\end{bmatrix}$

(B) $\begin{bmatrix}766 & -255 \\ 1530 & -509\end{bmatrix}$

(C) $\begin{bmatrix}1025 & -511 \\ 2024 & -1024\end{bmatrix}$

(D) $\begin{bmatrix}4 & -1 \\ 6 & -1\end{bmatrix}$

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Q27. MCQ (Single Correct Answer)

Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$, $\mathrm{QR}=5$ and the coordinates of the point P be $(0,2,3)$. If the area of the triangle PQR is $\frac{m}{n}$ then :

(A) $2\,\mathrm{m}-5\sqrt{21}\mathrm{n}=0$

(B) $\mathrm{m}-5\sqrt{21}\mathrm{n}=0$

(C) $5\,\mathrm{m}-21\sqrt{2}\mathrm{n}=0$

(D) $5\,\mathrm{m}-2\sqrt{21}\mathrm{n}=0$

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Q28. MCQ (Single Correct Answer)

Let $a \in R$ and A be a matrix of order 3×3 such that $\det(A)=-4$ and $A+I=\begin{bmatrix}1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2\end{bmatrix}$, where I is the identity matrix of order 3×3. If $\det((a+1)\,\mathrm{adj}((a-1)A))$ is $2^{\mathrm{m}}3^{\mathrm{n}}, \mathrm{m}, \mathrm{n} \in \{0,1,2,\ldots,20\}$, then $\mathrm{m}+\mathrm{n}$ is equal to :

(A) 14 (B) 17 (C) 15 (D) 16

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Q29. MCQ (Single Correct Answer)

If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2}\cos^2\theta+(2-\sqrt{6})\cos\theta-\sqrt{3}=0$, is equal to:

(A) 8 (B) 6 (C) 10 (D) 12

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Q30. MCQ (Single Correct Answer)

If S and S' are the foci of the ellipse $\frac{x^2}{18}+\frac{y^2}{9}=1$ and P be a point on the ellipse, then $\min(SP \cdot S'P)+\max(SP \cdot S'P)$ is equal to :

(A) $3(6+\sqrt{2})$ (B) $3(1+\sqrt{2})$ (C) 27 (D) 9

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Q31. MCQ (Single Correct Answer)

Let the focal chord PQ of the parabola $y^2 = 4x$ make an angle of 60° with the positive x axis, where P lies in the first quadrant. If the circle, whose one diameter is PS, S being the focus of the parabola, touches the y-axis at the point $(0, \alpha)$, then $5\alpha^2$ is equal to:

(A) 15 (B) 25 (C) 20 (D) 30

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Q32. MCQ (Single Correct Answer)

If the function $f(x) = 2x^3 - 9ax^2 + 12\mathrm{a}^2x + 1$, where $\mathrm{a} > 0$, attains its local maximum and local minimum values at p and q , respectively, such that $\mathrm{p}^2 = \mathrm{q}$, then $f(3)$ is equal to :

(A) 55 (B) 37 (C) 10 (D) 23

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Q33. MCQ (Single Correct Answer)

Let $ABCD$ be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the areas of the triangles ABC, ACD and ADB be 5,6 and 7 square units respectively. Then the area (in square units) of the $\triangle BCD$ is equal to :

(A) $\sqrt{110}$ (B) 12 (C) $\sqrt{340}$ (D) $7\sqrt{3}$

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Q34. MCQ (Single Correct Answer)

Let z be a complex number such that $|z| = 1$. If $\frac{2+\mathrm{k}^2 z}{\mathrm{k}+\bar{z}} = \mathrm{k}z, \mathrm{k} \in \mathbf{R}$, then the maximum distance of $\mathrm{k} + i\mathrm{k}^2$ from the circle $|z - (1 + 2i)| = 1$ is :

(A) $\sqrt{5} + 1$ (B) 3 (C) $\sqrt{3} + 1$ (D) 2

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Q35. MCQ (Single Correct Answer)

Let one focus of the hyperbola $\mathrm{H} : \frac{x^2}{\mathrm{a}^2} - \frac{y^2}{\mathrm{b}^2} = 1$ be at $(\sqrt{10}, 0)$ and the corresponding directrix be $x = \frac{9}{\sqrt{10}}$. If e and l respectively are the eccentricity and the length of the latus rectum of H , then $9\left(e^2 + l\right)$ is equal to :

(A) 12 (B) 14 (C) 15 (D) 16

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Q36. MCQ (Single Correct Answer)

The largest $\mathrm{n} \in \mathbf{N}$ such that 3^{n} divides 50 ! is :

(A) 22 (B) 20 (C) 21 (D) 23

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Q37. MCQ (Single Correct Answer)

Let $\mathrm{P}_{\mathrm{n}} = \alpha^{\mathrm{n}} + \beta^{\mathrm{n}}, \mathrm{n} \in \mathbf{N}$. If $\mathrm{P}_{10} = 123, \mathrm{P}_9 = 76, \mathrm{P}_8 = 47$ and $\mathrm{P}_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is :

(A) $x^2 + x - 1 = 0$ (B) $x^2 - x + 1 = 0$ (C) $x^2 + x + 1 = 0$ (D) $x^2 - x - 1 = 0$

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Q38. MCQ (Single Correct Answer)

If $\vec{a}$ is a nonzero vector such that its projections on the vectors $2\hat{i}-\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\vec{a}$ is :

(A) $\frac{1}{\sqrt{155}}(-7\hat{i}+9\hat{j}+5\hat{k})$

(B) $\frac{1}{\sqrt{155}}(-7\hat{i}+9\hat{j}-5\hat{k})$

(C) $\frac{1}{\sqrt{155}}(7\hat{i}+9\hat{j}-5\hat{k})$

(D) $\frac{1}{\sqrt{155}}(7\hat{i}+9\hat{j}+5\hat{k})$

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Q39. MCQ (Single Correct Answer)

If the system of linear equations

$3x+y+\beta z=3$
$2x+\alpha y-z=-3$
$x+2y+z=4$

has infinitely many solutions, then the value of $22\beta-9\alpha$ is :

(A) 31 (B) 37 (C) 43 (D) 49

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Q40. MCQ (Single Correct Answer)

Let $f:\mathbf{R}\to\mathbf{R}$ be a twice differentiable function such that $(\sin x\cos y)(f(2x+2y)-f(2x-2y))=(\cos x\sin y)(f(2x+2y)+f(2x-2y))$, for all $x,y\in\mathbf{R}$. If $f'(0)=\frac{1}{2}$, then the value of $24f''\left(\frac{5\pi}{3}\right)$ is :

(A) 2 (B) 3 (C) -3 (D) -2

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Q41. MCQ (Single Correct Answer)

Let A be the set of all functions $f:\mathbf{Z}\to\mathbf{Z}$ and R be a relation on A such that $\mathrm{R}=\{(\mathrm{f},\mathrm{g}):f(0)=\mathrm{g}(1)$ and $f(1)=\mathrm{g}(0)\}$. Then R is :

(A) Symmetric and transitive but not reflexive

(B) Symmetric but neither reflexive nor transitive

(C) Transitive but neither reflexive nor symmetric

(D) Reflexive but neither symmetric nor transitive

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Q42. MCQ (Single Correct Answer)

The number of sequences of ten terms, whose terms are either 0 or 1 or 2 , that contain exactly five 1 s and exactly three 2 s , is equal to :

(A) 360 (B) 2520 (C) 1820 (D) 45

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Q43. MCQ (Single Correct Answer)

For $\alpha, \beta, \gamma \in \mathbf{R}$, if $\lim\limits_{x \to 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$, then $\beta + \gamma - \alpha$ is equal to :

(A) -1 (B) 4 (C) 6 (D) 7

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Q44. MCQ (Single Correct Answer)

The term independent of x in the expansion of $\left(\frac{(x+1)}{(x^{2/3}+1-x^{1/3})} - \frac{(x-1)}{(x-x^{1/2})}\right)^{10}, x > 1$, is :

(A) 240 (B) 120 (C) 150 (D) 210

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Q45. MCQ (Single Correct Answer)

Let $a_1, a_2, a_3, \ldots$ be in an A.P. such that $\sum\limits_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$. If $\sum\limits_{k=1}^{n} a_k = 0$, then n is :

(A) 18 (B) 17 (C) 11 (D) 10

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Q46. Numerical

Three distinct numbers are selected randomly from the set $\{1, 2, 3, \ldots, 40\}$. If the probability, that the selected numbers are in an increasing G.P., is $\frac{m}{n}, \gcd(m, n) = 1$, then $m + n$ is equal to _____ .

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Q47. Numerical

Let $f : \mathbf{R} \to \mathbf{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0, f''(x) = f(x), f(0) = 0, f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____ .

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Q48. Numerical

If the area of the region $\{(x, y) : |4 - x^2| \leq y \leq x^2, y \leq 4, x \geq 0\}$ is $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right), \alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to _____.

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Q49. Numerical

The absolute difference between the squares of the radii of the two circles passing through the point $(-9, 4)$ and touching the lines $x + y = 3$ and $x - y = 3$, is equal to _____ .

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Q50. Numerical

Let [.] denote the greatest integer function. If $\int_0^{e^3} \left[\frac{1}{e^x-1}\right] dx = \alpha - \log_e 2$, then α^3 is equal to _____.

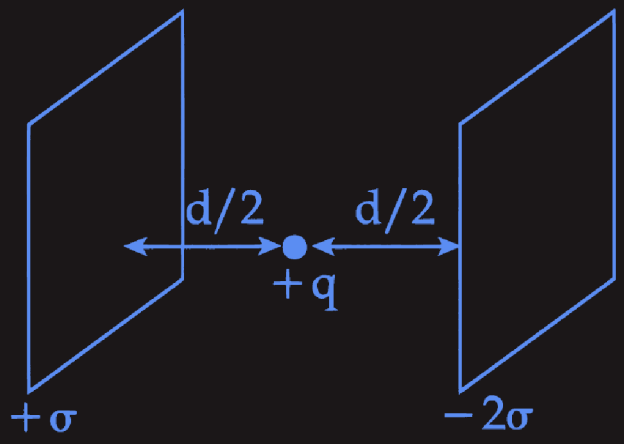
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Physics 25 questions

Q51. MCQ (Single Correct Answer)

Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ. The force experienced by a point charge +q placed at the mid point between two plates will be:

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(A) $\frac{3\sigma q}{4\epsilon_0}$

(B) $\frac{3\sigma q}{2\epsilon_0}$

(C) $\frac{\sigma q}{4\epsilon_0}$

(D) $\frac{\sigma q}{2\epsilon_0}$

Q52. MCQ (Single Correct Answer)

A monochromatic light is incident on a metallic plate having work function ϕ. An electron, emitted normally to the plate from a point A with maximum kinetic energy, enters a constant magnetic field, perpendicular to the initial velocity of electron. The electron passes through a curve and hits back the plate at a point B. The distance between A and B is : (Given : The magnitude of charge of an electron is e and mass is m, h is Planck's constant and c is velocity of light. Take the magnetic field exists throughout the path of electron)

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(A) $\sqrt{2m\left(\frac{hc}{\lambda}-\phi\right)}/\mathrm{eB}$

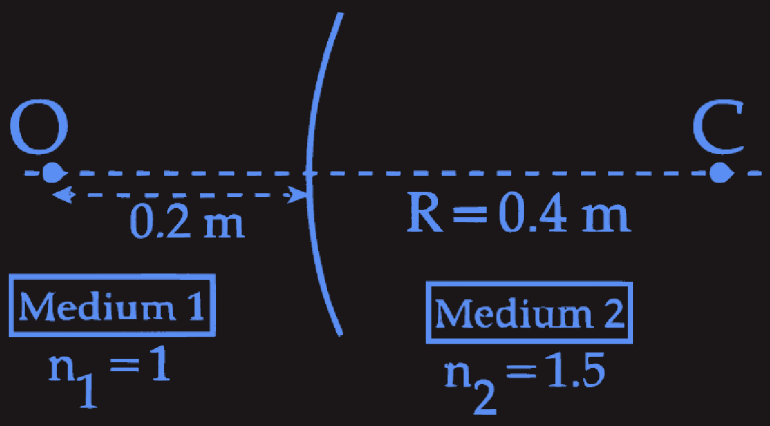
(B) $\sqrt{8\,\mathrm{m}\left(\frac{\mathrm{hc}}{\lambda}-\phi\right)}/\mathrm{eB}$

(C) $\sqrt{\mathrm{m}(\mathrm{hc}/\lambda-\phi)}/\mathrm{eB}$

(D) $2\sqrt{\mathrm{m}(\mathrm{hc}/\lambda-\phi)}/\mathrm{eB}$

Q53. MCQ (Single Correct Answer)

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A spherical surface separates two media of refractive indices 1 and 1.5 as shown in figure. Distance of the image of an object ' O ', is :

(C is the center of curvature of the spherical surface and R is the radius of curvature)

(A) 0.24 m left to the spherical surface
(B) 0.24 m right to the spherical surface
(C) 0.4 m left to the spherical surface
(D) 0.4 m right to the spherical surface

Q54. MCQ (Single Correct Answer)

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A river is flowing from west to east direction with speed of $9\ \mathrm{km\ h^{-1}}$. If a boat capable of moving at a maximum speed of $27\ \mathrm{km\ h^{-1}}$ in still water, crosses the river in half a minute, while moving with maximum speed at an angle of 150° to direction of river flow, then the width of the river is :

(A) 112.5 m
(B) 75 m
(C) 300 m
(D) $112.5 \times \sqrt{3}\ \mathrm{m}$

Q55. MCQ (Single Correct Answer)

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Moment of inertia of a rod of mass ' M ' and length ' L ' about an axis passing through its center and normal to its length is ' α '. Now the rod is cut into two equal parts and these parts are joined symmetrically to form a cross shape. Moment of inertia of cross about an axis passing through its center and normal to plane containing cross is :

(A) $\alpha/4$
(B) $\alpha/8$
(C) α
(D) $\alpha/2$

Q56. MCQ (Single Correct Answer)

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Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I. Let $\mathrm{B_2}$ be the magnitude of magnetic field at an axial distance ' x ' from the center. For $x : \mathrm{R} = 3 : 4$, $\frac{\mathrm{B_2}}{\mathrm{B_1}}$ is :

(A) $64 : 125$
(B) $25 : 16$
(C) $4 : 5$
(D) $16 : 25$

Q57. MCQ (Single Correct Answer)

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A light wave is propagating with plane wave fronts of the type $x + y + z =$ constant. The angle made by the direction of wave propagation with the x-axis is :

(A) $\cos^{-1}(2/3)$

(B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) $\cos^{-1}\left(\frac{1}{3}\right)$

(D) $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Q58. MCQ (Single Correct Answer)

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Match List I with List II.

	List - I		List - II
(A)	Coefficient of viscosity	(I)	$\left[\mathrm{ML^0\,T^{-3}}\right]$
(B)	Intensity of wave	(II)	$\left[\mathrm{ML^{-2}\,T^{-2}}\right]$
(C)	Pressure gradient	(III)	$\left[\mathrm{M^{-1}LT^2}\right]$
(D)	Compressibility	(IV)	$\left[\mathrm{ML^{-1}\,T^{-1}}\right]$

Choose the correct answer from the options given below:

(A) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(B) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(D) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Q59. MCQ (Single Correct Answer)

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A particle is subjected to two simple harmonic motions as : $x_1 = \sqrt{7}\sin 5t$ cm and $x_2 = 2\sqrt{7}\sin\left(5t + \frac{\pi}{3}\right)$cm where x is displacement and t is time in seconds. The maximum acceleration of the particle is $x \times 10^{-2}\ \mathrm{ms}^{-2}$. The value of x is :

(A) $5\sqrt{7}$

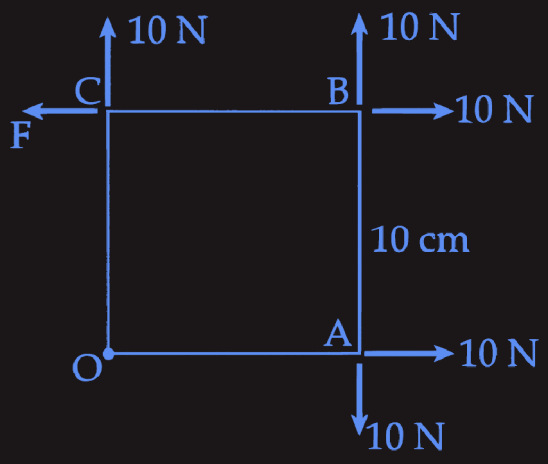
(B) 125

(C) $25\sqrt{7}$

(D) 175

Q60. MCQ (Single Correct Answer)

A square Lamina OABC of length 10 cm is pivoted at ' O '. Forces act at Lamina as shown in figure. If Lamina remains stationary, then the magnitude of F is :



(A) 10 N (B) 0 (zero) (C) $10\sqrt{2}$ N (D) 20 N

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Q61. MCQ (Single Correct Answer)

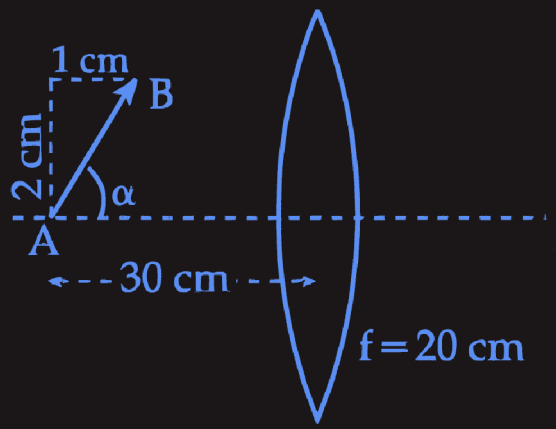
A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(\mathrm{d}, 0, 0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is:

(A) $(3\mathrm{d}/4, 0, 0)$ (B) $(\mathrm{d}/4, 0, 0)$ (C) $(4\mathrm{d}/3, 0, 0)$ (D) $(\mathrm{d}/3, 0, 0)$

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Q62. MCQ (Single Correct Answer)

A slanted object AB is placed on one side of convex lens as shown in the diagram. The image is formed on the opposite side. Angle made by the image with principal axis is :



(A) $-\alpha$ (B) $-\frac{\alpha}{2}$ (C) -45° (D) $+45^\circ$

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Q63. MCQ (Single Correct Answer)

Considering Bohr's atomic model for hydrogen atom :

(A) the energy of H atom in ground state is same as energy of He^{+} ion in its first excited state.

(B) the energy of H atom in ground state is same as that for Li^{++} ion in its second excited state.

(C) the energy of H atom in its ground state is same as that of He^{+} ion for its ground state.

(D) the energy of He^{+} ion in its first excited state is same as that for Li^{++} ion in its ground state.

Choose the correct answer from the options given below :

(A) (A), (B) only (B) (A), (D) only (C) (A), (C) only (D) (B), (D) only

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Q64. MCQ (Single Correct Answer)

In an adiabatic process, which of the following statements is true?

(A) The internal energy of the gas decreases as the temperature increases

(B) The molar heat capacity is zero

(C) Work done by the gas equals the increase in internal energy

(D) The molar heat capacity is infinite

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Q65. MCQ (Single Correct Answer)

The equation for real gas is given by $\left(\mathrm{P}+\frac{\mathrm{a}}{\mathrm{V}^2}\right)(\mathrm{V}-\mathrm{b})=\mathrm{RT}$, where $\mathrm{P}, \mathrm{V}, \mathrm{T}$ and R are the pressure, volume, temperature and gas constant, respectively. The dimension of ab^{-2} is equivalent to that of :

(A) Compressibility (B) Planck's constant

(C) Energy density (D) Strain

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Q66. MCQ (Single Correct Answer)

The relationship between the magnetic susceptibility (χ) and the magnetic permeability (μ) is given by :

(μ_0 is the permeability of free space and μ_r is relative permeability)

(A) $\chi=\frac{\mu}{\mu_0}-1$ (B) $\chi=\frac{\mu_r}{\mu_0}+1$

(C) $\chi=\mu_r+1$ (D) $\chi=1-\frac{\mu}{\mu_0}$

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Q67. MCQ (Single Correct Answer)

The battery of a mobile phone is rated as $4.2\ \mathrm{V}, 5800\ \mathrm{mAh}$. How much energy is stored in it when fully charged?

(A) 87.7 kJ (B) 43.8 kJ (C) 48.7 kJ (D) 24.4 kJ

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Q68. MCQ (Single Correct Answer)

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A zener diode with 5 V zener voltage is used to regulate an unregulated dc voltage input of 25 V . For a 400Ω resistor connected in series, the zener current is found to be 4 times load current. The load current (I_L) and load resistance (R_L) are :

(A) $\mathrm{I_L = 0.02\,mA; R_L = 250\Omega}$

(B) $\mathrm{I_L = 10\,A; R_L = 0.5\Omega}$

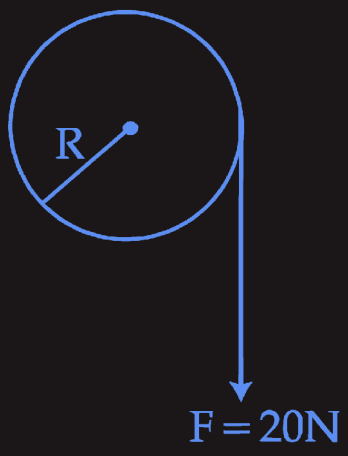
(C) $\mathrm{I_L = 10\,mA; R_L = 500\Omega}$

(D) $\mathrm{I_L = 20\,mA; R_L = 250\Omega}$

Q69. MCQ (Single Correct Answer)

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A cord of negligible mass is wound around the rim of a wheel supported by spokes with negligible mass. The mass of wheel is 10 kg and radius is 10 cm and it can freely rotate without any friction. Initially the wheel is at rest. If a steady pull of 20 N is applied on the cord, the angular velocity of the wheel, after the cord is unwound by 1 m , would be:



(A) 20 rad/s

(B) 30 rad/s

(C) 10 rad/s

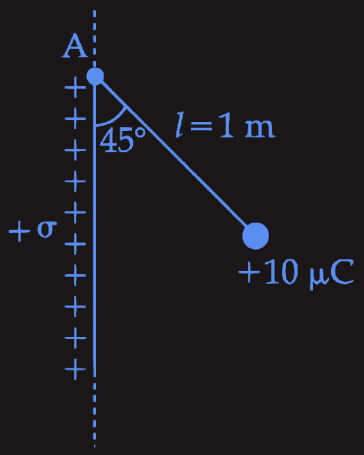
(D) 0 rad/s

Q70. MCQ (Single Correct Answer)

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A small bob of mass 100 mg and charge $+10\mu C$ is connected to an insulating string of length 1 m . It is brought near to an infinitely long non-conducting sheet of charge density ' σ ' as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be.

(Given, $\epsilon_0 = 8.85 \times 10^{-12} \frac{\mathrm{F}}{\mathrm{m}}$ and acceleration due to gravity, $\mathrm{g} = 10 \frac{\mathrm{m}}{\mathrm{s}^2}$)



(A) $1.77\ \mathrm{nC/m^2}$

(B) $0.885\ \mathrm{nC/m^2}$

(C) $885\ \mathrm{nC/m^2}$

(D) $17.7\ \mathrm{nC/m^2}$

Q71. Numerical

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A steel wire of length 2 m and Young's modulus $2.0 \times 10^{11}\ \mathrm{N\,m^{-2}}$ is stretched by a force. If Poisson ratio and transverse strain for the wire are 0.2 and 10^{-3} respectively, then the elastic potential energy density of the wire is _____ $\times 10^5$ (in SI units).

Q72. Numerical

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γ_A is the specific heat ratio of monoatomic gas A having 3 translational degrees of freedom. γ_B is the specific heat ratio of polyatomic gas B having 3 translational, 3 rotational degrees of freedom and 1 vibrational mode. If $\frac{\gamma_A}{\gamma_B} = \left(1 + \frac{1}{n}\right)$, then the value of n is _____ .

Q73. Numerical

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A vessel with square cross-section and height of 6 m is vertically partitioned. A small window of $100\ \mathrm{cm^2}$ with hinged door is fitted at a depth of 3 m in the partition wall. One part of the vessel is filled completely with water and the other side is filled with the liquid having density $1.5 \times 10^3\ \mathrm{kg/m^3}$. What force one needs to apply on the hinged door so that it does not get opened ?

(Acceleration due to gravity $= 10\ \mathrm{m/s^2}$)

Q74. Numerical

If the measured angular separation between the second minimum to the left of the central maximum and the third minimum to the right of the central maximum is 30° in a single slit diffraction pattern recorded using 628 nm light, then the width of the slit is _____ μm.

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Q75. Numerical

A person travelling on a straight line moves with a uniform velocity v_1 for a distance x and with a uniform velocity v_2 for the next $\frac{3}{2}x$ distance. The average velocity in this motion is $\frac{50}{7}$ m/s. If v_1 is 5 m/s then $v_2 =$ _____ m/s.

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Answer Key

Chemistry

1. **D**	2. **C**	3. **B**	4. **C**	5. **A**
6. **B**	7. **C**	8. **B**	9. **D**	10. **C**
11. **A**	12. **C**	13. **B**	14. **D**	15. **B**
16. **C**	17. **B**	18. **C**	19. **C**	20. **A**
21. **2435**	22. **372**	23. **6**	24. **74**	25. **1**

Mathematics

26. **B**	27. **A**	28. **D**	29. **A**	30. **C**
31. **A**	32. **B**	33. **A**	34. **A**	35. **D**
36. **A**	37. **A**	38. **D**	39. **A**	40. **C**
41. **B**	42. **B**	43. **D**	44. **D**	45. **C**
46. **2477**	47. **36**	48. **22**	49. **768**	50. **8**

Physics

51. **B**	52. **B**	53. **C**	54. **A**	55. **A**
56. **A**	57. **B**	58. **C**	59. **D**	60. **A**
61. **B**	62. **C**	63. **A**	64. **B**	65. **C**
66. **A**	67. **A**	68. **C**	69. **A**	70. **A**
71. **25**	72. **3**	73. **150**	74. **6**	75. **10**

For numerical answers given as a range, any value within the range is correct.

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